

Taxonomy of Multimodal Time Series Models

Multimodal Time Series Paradigm (Data, Fusion, Role, Task, Domain)

1. Modality Pairing

- TS + Text (Reports, News, Prompts)
- TS + Vision (Plots, Spectrograms, MAE)
- TS + Audio (Speech AMs, Bioacoustics)
- TS + Vision + Text (Tri-modal VLM, EHR)
- TS + Physics / Spatio-Temporal Fields
 - Omnimodal / Multi-sensor Data

2. Fusion Architecture

- Patch Reprogramming (Time-LLM, OFA)
- Acoustic Reprogramming (Voice2Series)
- Visual Rendering (VisionTS, VisionTS++)
 - Variable-Agnostic ViT (ClimaX, Prithvi)
- Cross-Attention Adapter (Time-VLM, MedFuse)
- Unified Early Tokenization (UniTS, ChatTS)

3. Non-TS Role

- Auxiliary Context / Condition
 - Frozen Transfer Substrate
- Conversational & Reasoning Interface
- Metric Alignment Target (Retrieval)
 - Multi-Task Instruction Steering
- Tool & Neuro-Symbolic Querying

4. Downstream Tasks

- Multimodal Forecasting (Point/Prob.)
- Anomaly Detection & Phase Picking
 - Time Series QA & Reasoning
- Cross-Modal Retrieval (TS <-> Text)
- Clinical Risk & Mortality Phenotyping
- Earth System Weather / Downscaling

Application Domains: Healthcare & EHR • Meteorology & Climate • Finance & Markets • Energy & Smart Grids • Geophysics & Bioacoustics